

SIMATS ENGINEERING
ASSESSMENT TOOL 3: MCQ Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1596

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Dave's Taxonomy: Precision (Level 3)

Total Marks: 30

Question Count: 15

Duration : 60 Minutes

Code of Conduct

I NITHISH KUMAR K(192521266) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: knithishkumar

MCQ Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Understanding	30%	Answers reflect strong and accurate concept mastery	Mostly accurate understanding	Partial understanding	Weak understanding
Accuracy of Responses	30%	Nearly all responses are correct	Most responses are correct	Some responses are correct	Few responses are correct
Application Awareness	15%	Shows strong recognition of practical cloud use cases	Reasonable recognition	Limited recognition	Weak recognition
Time Management	10%	Completes assessment efficiently	Completes on time	Needs extra time	Unable to finish well
Question Interpretation	15%	Interprets all items correctly	Misreads very few items	Misreads several items	Frequently misinterprets questions

Total Marks Awarded:

Assessment Tasks

MCQ Test

- 1. A company's cloud storage system requires 5 TB of data replication across three regions. What is the total storage required?**
 - a) 5 TB
 - b) 10 TB
 - c) 15 TB
 - d) 20 TB
- 2. A cloud storage provider offers a 99.99% uptime SLA. How much downtime can be expected per month?**
 - a) ~4 minutes
 - b) ~40 minutes
 - c) ~4 hours
 - d) ~24 hours
- 3. A cloud-based load balancer distributes 60,000 requests per second among 10 servers. What is the average number of requests per server per second?**
 - a) 5000
 - b) 6000
 - c) 7000
 - d) 10,000
- 4. A company's cloud network has 1 Gbps bandwidth and needs to transfer 500 GB of data. How long will it take?**
 - a) 1 hour
 - b) 10 hours
 - c) 1.1 hours
 - d) 5.5 hours
- 5. A cloud storage system experiences a 99.95% availability rate. How much downtime does it experience per year?**
 - a) ~4 hours
 - b) ~8 hours
 - c) ~20 hours
 - d) ~40 hours
- 6. A cloud provider offers pay-as-you-go pricing at \$0.05 per GB for storage. If a company stores 10 TB, what is the monthly cost?**
 - a) \$50

- b) \$500
- c) \$1000
- d) \$2000

7. A cloud network has 10 Gbps throughput but experiences 2% packet loss. What is the effective throughput?

- a) 9.8 Gbps
- b) 9.6 Gbps
- c) 8.0 Gbps
- d) 7.5 Gbps

8. A user uploads 250 MB of data per day to the cloud. How much data will be uploaded in one year (365 days)?

- a) 91.25 GB
- b) 85 GB
- c) 100 GB
- d) 120 GB

9. A cloud-based video streaming service requires 5 Mbps bandwidth per user. If 100 users are streaming simultaneously, what is the total required bandwidth?

- a) 500 Mbps
- b) 250 Mbps
- c) 200 Mbps
- d) 1000 Mbps

10. A cloud provider guarantees 99.95% uptime per year. How much downtime does this allow in minutes per year?

- a) 262 minutes
- b) 365 minutes
- c) 525 minutes
- d) 150 minutes

11. A virtualized data center has 200 physical servers. If each server runs 40 virtual machines, what is the total number of VMs in the data center?

- a) 4,000
- b) 8,000
- c) 6,000
- d) 10,000

12. A physical server has 128 GB of RAM and hosts 10 virtual machines, each requiring 12 GB. How much memory is overcommitted?

- a. 4 GB
- b) 6 GB
- c) -8 GB
- d) 10 GB

13. A virtual machine requires 4 vCPUs. If a server has a total of 64 cores and an overcommitment ratio of 2:1, how many VMs can be hosted?

- a. 16
- b) 32
- c) 64
- d) 128

14. A company migrates 50 on-premise servers to the cloud, reducing power consumption by 60%. If the original power usage was 100 kW, what is the new power consumption?

- a. 40 kW
- b) 50 kW
- c) 60 kW
- d) 70 kW

15. A cloud provider operates a virtualized data center with 100 physical servers, each with 64 vCPUs. If the overcommitment ratio is 4:1, how many total vCPUs can be allocated?

- a. 6,400
- b) 12,800
- c) 25,600
- d) 50,000

16. Which cloud deployment model is used for organizations that require both private and public cloud resources?

- A): Hybrid Cloud
- B): Public Cloud
- C): Private Cloud
- D): Community Cloud

17. Which cloud security strategy ensures that sensitive data remains private even if intercepted?

- A): Encryption
- B): Load balancing
- C): Virtualization
- D): Resource pooling

18. If a hypervisor hosts 50 VMs and each VM uses 20 GB bandwidth per day, calculate total bandwidth consumption per week.

- a:** 5000 GB
- b:** 7000 GB
- c:** 6000 GB
- d:** 7500 GB

19. A cloud system runs at 70% average CPU utilization. If it has 32 cores at 2.5 GHz each, what is total utilized processing power?

- a:** 56 GHz
- b:** 58 GHz
- c:** 64 GHz
- d:** 70 GHz

20. If the network latency in a virtualized system is 5 ms and processing time is 10 ms, calculate total round-trip delay for 100 requests.

- a:** 1.5 sec
- b:** 2 sec
- c:** 1 sec
- d:** 3 sec

21. If 100 VMs transfer logs of 200 MB/day each to the cloud, calculate the total data transferred in a 30-day month.

- a:** 600 GB
- b:** 500 GB
- c:** 700 GB
- d:** 800 GB

22. A cloud service's availability is 99.95%. How much downtime (in minutes) is expected in a month (30 days)?

- a:** 21.6 min
- b:** 36 min
- c:** 43.2 min
- d:** 18 min

23. If a cloud server handles 200 requests/sec and each request uses 10 KB memory buffer, compute memory usage per minute.

- a:** 120 MB
- b:** 100 MB
- c:** 150 MB
- d:** 200 MB

24. A hypervisor hosts 120 VMs. If 75% of them require 2 vCPUs each, calculate total vCPU requirement.

- a:** 180
- b:** 150
- c:** 120
- d:** 160

25. A company has 20 racks with 20 servers each. If each server uses 3 virtual NICs, how many vNICs exist in total?

- a:** 1200
- b:** 1000
- c:** 800
- d:** 600

26. If a hypervisor can boot a VM in 2 minutes, how long will it take to boot 60 VMs in batches of 10 concurrently?

- a:** 12 min
- b:** 10 min
- c:** 6 min
- d:** 8 min

27. A cloud data center uses 5 cooling units, each consuming 3.5 kW. Find total cooling energy per day.

Formula:

$\text{Energy} = \text{Units} \times \text{Power} \times \text{Time (hrs)}$

- a:** 420 kWh
- b:** 480 kWh
- c:** 400 kWh
- d:** 500 kWh

28. Each server is configured with 2 TB storage. A virtual SAN clusters 20 such servers. What is total usable capacity, assuming 20% is used for redundancy?

- a: 32 TB
- b: 35 TB
- c: 30 TB
- d: 28 TB

29. Which of the following is NOT a benefit of virtualization?

- i. Reduced hardware costs
- b) Increased energy consumption
- c) Improved resource management
- d) Faster deployment of applications

30. Which feature in virtualization allows multiple OS environments to run simultaneously?

- i. Hypervisor
- b) Load balancer
- c) Firewall
- d) BIOS

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cloud computing

MCQ Test

01. cloud storage replication:

☒ a) 1 TB.

29
20

02. 99.99% uptime SLA.

☒ a) ~4 minutes.

03. Load balancer.

☒ b) 6000

04. 1 Gbps, 500 GB transfer

☒ c) 11 hours.

05. 99.95% availability per year

☒ a) ~4 hours

06. 10 TB storage at \$0.05 / GB

☒ b) \$500

07. 10 Gbps with 2% packet loss

☒ a) 9.8 Gbps.

08. 250 MB/day for 365 days

☒ a) 91.25 GB.

09. 5 Mbps x 100 users

☒ a) 500 Mbps

10. 99.95% uptime / per year

c) 525 minutes.

11. 200 servers x 40 VMs

b) 8000 VM's

12. 128 GiB RAM, 10 x 12 GiB VMs

a) 4 GiB.

13. 4 vCPU VM, 64 cores, 2:1 ratio

c) 64 VM's

14. 60% power reduction from 100 kW

a) 40 kW.

15. 100 x 64 vCPU, 4:1 ratio

c) 25,600 vCPUs

16. private + public cloud

A) Hybrid cloud.

17. sensitive data protection

A) Encryption.

18. 50 VMs x 20 GiB / day x 7 days

b) 7000 GiB.

20. $5\text{ms} + 10\text{ms} \times 100 \text{ requests}$

a) 1.5 sec

21. $100 \text{ VMs} \times 200 \text{ MB/day} \times 30 \text{ days}$

a) 600 GB

22. 99.95% availability / month

a) 21.6 min.

23. $200 \text{ request/sec} \times 10 \text{ KB} \times 60 \text{ sec.}$

a) 120 MB

24. 75% of 120 VMs $\times 2 \text{ vCPU}$

a) 180 vCPU

25. 20 racks $\times 20 \text{ servers} \times 3 \text{ vNIC's}$

a) 1200

26. 60 VMs, batches of 10, 3 min / batch

a) 12 min.

27. 5 cooling units $\times 3.5 \text{ kW} \times 24 \text{ hrs}$

a) 420 kWh.

28. 20 servers $\times 2 \text{ TB}$, 20% redundancy.

a) 32 TB

29. NOT a virtualization benefit

increased energy consumption.

~~20. 47 Hypervisor~~

30. b) Load balancer